

AI Small Business Operations Agent

Assignment 2 - AI Agent Project

1. Problem Statement

Small businesses often manage customer orders, inventory, customer messages, and operational decisions **manually**. This causes delayed customer responses, stock mistakes and overselling, unnecessary manual work, and inconsistent decision-making. There is no affordable AI-powered solution that can automatically process orders, check inventory, make intelligent decisions, and notify both customers and business owners in real time.

2. Proposed AI Agent Solution

An AI-powered operations agent that receives customer order requests, understands intent using Deepseek LLM, checks inventory in Google Sheets, makes intelligent decisions based on business rules, and notifies customers and the business owner via Telegram. The agent autonomously routes orders through 5 different paths based on context: approve, escalate, suggest alternative, clarify, or reject.

3. Workflow Diagram

Customer sends order/request → AI extracts intent (Decision 1) → Check inventory in Google Sheets → AI decides action (Decisions 2 & 3) → Route based on action (autonomous workflow-changing decision) → Execute action (update sheets, send notifications) → Log the action → Return result

Step	Action	Tool/Service
1	Receive customer message	Webhook / CLI / Web
2	AI extracts order intent	Deepseek LLM (Decision 1)
3	Check inventory	Google Sheets API
4	AI decides action	Deepseek LLM (Decisions 2 & 3)
5	Route based on action	Autonomous routing (5 paths)
6	Update inventory & orders	Google Sheets API
7	Notify customer	Telegram Bot API
8	Notify owner (if escalated)	Telegram Bot API
9	Log the action	Google Sheets API
10	Return result	Webhook response

4. System Architecture

The system consists of a Python application (agent.py) that orchestrates the workflow. It connects to three external services: Deepseek API for LLM reasoning, Google Sheets API for data storage, and Telegram Bot API for notifications. The agent can be run in CLI mode, web mode (Flask), or demo mode.

Component	Technology	Purpose
Orchestration	Python (agent.py)	Workflow engine, routing, data flow
LLM	Deepseek (deepseek-chat)	Intent extraction + decision making
Database	Google Sheets	Products, Orders, Customers, Logs
Notifications	Telegram Bot API	Customer + owner messaging
Input	CLI / Web / Webhook	Customer order entry point

5. AI Tools, Models, and External Services

Service	Type	Purpose
Deepseek API	LLM (deepseek-chat)	Intent extraction, decision making, customer responses
Google Sheets API	External Tool	Inventory management, order records, customer data, audit logs
Telegram Bot API	External Tool	Real-time notifications to customers and business owner
Python (agent.py)	Framework	Workflow orchestration, routing, data processing
Flask (web_app.py)	Web Framework	Web interface for testing and demo

6. AI Decision Points

- Decision 1 — Extract Order Intent:** The LLM parses the customer message to identify product, variant, quantity, and intent.
- Decision 2 — Can the order be fulfilled?** The LLM compares requested quantity against available stock.
- Decision 3 — What action should be taken?** The LLM applies business rules to select the best action.

Decision	Input	Output
1. Extract Intent	Customer message	Product, variant, quantity, intent
2. Fulfillment Check	Order + inventory	Can/cannot fulfill
3. Action Selection	Rules + context	approve / escalate / suggest / clarify / reject

7. Autonomous Workflow-Changing Decision

The agent autonomously selects one of 5 paths based on the AI decision. This is the workflow-changing decision that demonstrates the agent's autonomy:

- **Approve:** Auto-fulfill order, update stock, notify customer
- **Escalate:** Notify owner for manual approval
- **Suggest Alternative:** Offer substitute products
- **Clarify:** Ask customer for more information
- **Reject:** Decline the order

8. Sample Execution

Input: "I want 2 black Basic Shirts"

Output:

```
{"status": "approved", "order_id": "ORD-2CED1883", "total": 20}
```

The agent extracted the intent (Basic Shirt, Black, qty 2), checked inventory (20 in stock), decided to auto-approve (total \$20 < \$100), updated stock (20→18), created an order record, notified the customer via Telegram, and logged the action.

9. Requirements Verification

Requirement	Status	Evidence
Defined real-world problem	Met	Small business order management
LLM for reasoning/decisions	Met	Deepseek API (deepseek-chat)
≥2 external tools/services	Met	Google Sheets API, Telegram Bot API
≥5 workflow steps	Met	10 steps
≥2 AI decision points	Met	3 decisions
≥1 autonomous decision	Met	5-way routing

10. Project Link

The project is a standalone Python application. The main agent is in **agent.py**. The web interface is in **web_app.py**. The demo script is **demo.py**. All configuration is in **.env**.

How to run:

```
python agent.py    # Interactive CLI mode
```

```
python web_app.py    # Web interface (http://localhost:5000)
```

```
python demo.py      # Run all test scenarios
```